BRD-Lite: E-Commerce Web Application (Amazon-like)

1. Objective

Define requirements for a web-based e-commerce platform that allows users to:

- Browse products
- Add to cart
- Purchase items
- Track orders

Goal: Ensure seamless shopping experience with secure transactions.

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2. Scope

In Scope

- User Registration & Login
- Product Search & Listing
- Product Details Page
- Cart Management
- Checkout & Payment
- Order History & Tracking

Out of Scope (important — most people skip this)

- Seller onboarding
- Warehouse/logistics management
- Advanced recommendation engine

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3. User Roles

Role	Description
Guest User	Can browse/search products
Registered User	Can purchase, track orders
Admin (optional)	Manage products/orders

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4. Functional Requirements

4.1 User Authentication

- User can register using email/mobile
- User can login/logout
- Invalid credentials show error

👉 Edge cases you should test:

- Empty fields
- Invalid email format
- Multiple failed login attempts

4.2 Product Search & Listing

- User can search products via keyword
- Filters: price, category, rating
- Sort: low to high, high to low

👉 Critical QA angle:

- Search relevance
- No-result scenarios
- Special characters

4.3 Product Details Page (PDP)

- Display:
  - Product name
  - Price
  - Description
  - Ratings
- “Add to Cart” button

👉 Test focus:

- Missing data
- Incorrect price display
- Image loading failures

4.4 Cart Management

- Add product to cart
- Update quantity
- Remove item
- View total price

👉 Critical validations:

- Price calculation accuracy
- Duplicate items handling
- Max quantity limits

4.5 Checkout Process

- Enter shipping address
- Select payment method
- Place order

👉 High-risk area:

- Address validation
- Payment failure handling
- Session timeout during checkout

4.6 Payment

- Modes: Card, UPI, COD
- Payment success/failure response

👉 Must test:

- Payment retry
- Duplicate transactions
- Timeout scenarios

4.7 Order Management

- Order confirmation page
- Order history
- Order tracking

👉 QA focus:

- Order ID generation
- Status updates (Placed → Shipped → Delivered)
- Data consistency

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5. Non-Functional Requirements

Type	Requirement
Performance	Pages load < 3 sec
Security	Secure payment (HTTPS, encryption)
Usability	Simple navigation
Compatibility	Works across browsers/devices

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6. Assumptions

- Stable internet connection
- Third-party payment gateway works correctly
- Product inventory is pre-managed

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7. Dependencies

- Payment gateway API
- Product database
- Authentication service

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8. Risks

Risk	Impact
Payment failure	Revenue loss
Server downtime	User drop
Incorrect pricing	Customer dissatisfaction

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9. Success Criteria

- User completes purchase without errors

- Accurate order processing
- No critical defects in checkout/payment